Xilong Wang

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Education

Duke University Durham, NC
Ph.D. student in ECE Aug. 2024 – Present
Advisor: Prof. Neil Gong

University of Science and Technology of China (USTC) Hefei, China
Undergraduate in Cyber Science and Technology Sept. 2020 – June. 2024

Research Interests

LLM Agent Security, Prompt Injection Attacks and Defenses, Trustworthy AI

Experience

Research Intern, Li Auto Mar. 2024 – Jul. 2024
Research topic: LLM Reasoning and String Processing
Mentor: Dr. Hao Fu

- Developed StringBench to evaluate the string processing capabilities of LLMs and identify their limitations, and proposed a fine-tuning-based method to address these limitations, resulting in a paper published at ICLR 2025.

Research Intern, Purdue University Jul. 2023 – Nov. 2023
Research topic: Prompt Injection Attacks
Advisor: Prof. Xiangyu Zhang

Undergraduate Researcher, Duke University Aug. 2022 – Oct. 2023
Research topic: Federated Learning Security
Advisor: Prof. Neil Gong

Undergraduate Researcher, University of Science and Technology of China Apr. 2021 – Oct. 2022
Research topic: Linguistic Steganography
Advisor: Prof. Weiming Zhang

Publications

(* Equal contribution) (‡ Students mentored by me)

Same-Origin Policy for Agentic Browsers. PDF Code
X. Wang, X. Chen‡, P. Li, D. Song, and N. Gong

CleanBase: Detecting Malicious Documents in RAG Knowledge Databases. PDF Code
W. Jin‡*, **X. Wang***, W. Zou, J. Jia, and N. Gong

AlignSentinel: Alignment-Aware Detection of Prompt Injection Attacks. PDF
Y. Jia, R. Wang, **X. Wang**, C. Xiang, and N. Gong

WebSentinel: Detecting and Localizing Prompt Injection Attacks for Web Agents. PDF Code
X. Wang, Y. Liu[‡], Z. Wang, D. Song, and N. Gong

WAInjectBench: Benchmarking Prompt Injection Detections for Web Agents. PDF Code
Y. Liu[‡*], **X. Wang**[*], R. Xu[‡], Y. Jia, and N. Gong
ACM CCS Workshop on Security, Privacy, and Safety of Agentic AI Systems, 2026

WebInject: Prompt Injection Attack to Web Agents. PDF
X. Wang, J. Bloch[‡], Z. Shao, Y. Hu, S. Zhou, and N. Gong
EMNLP, 2025.

StringLLM: Understanding the String Processing Capability of Large Language Models. PDF Code
X. Wang, H. Fu, J. Wang, and N. Gong
ICLR, 2025.

Provably Robust Federated Reinforcement Learning. PDF
M. Fang[*], **X. Wang**[*], and N. Gong
WWW **(Oral)**, 2025.

ICStega: Image Captioning-based Semantically Controllable Linguistic Steganography. PDF
X. Wang, Y. Wang, K. Chen, J. Ding, W. Zhang, and N. Yu.
ICASSP, 2023

Awards

Duke ECE Departmental Fellowship 2024
USTC Outstanding Student Scholarship (Top 5%), First Prize 2021
USTC Outstanding Student Scholarship (Top 5%), First Prize 2020

Academic Activities

Reviewer
ICLR 2026, ICML 2026, NeurIPS 2026, IEEE T-IFS

Teaching Experience

Teaching Assistant, Duke University Spring 2026
Generative AI: Foundations, Applications, and Safety, ECE590

Teaching Assistant, USTC Fall 2023
Design and Practice of Information Security II